Proof of Concept

## 1. Primary Objective of the Algorithmic Proof

### Core hypothesis to prove

> Using the Samsung Galaxy Watch5 Pro, we can derive a usable stress/HRV signal and detect movement states—rest, walking, tremor-like motion, and gait irregularity—well enough to trigger personalized haptic cues.

### What you are not trying to prove yet

Do **not** try to prove any of these yet:

- That the watch treats Parkinson's disease.
- That vibration reduces tremor.
- That the app prevents falls.
- That it improves speech.
- That it is medically safe or clinically validated.

For now, the product is a **research/wellness data collection and cueing prototype**.

## 2. Define the MVP Algorithmic Proof

Your first proof should answer five questions.

### Question 1: Signal quality

Can we extract reliable heart rate and HRV-related features from the watch during:

- Rest
- Stress
- Walking
- Arm movement
- Tremor-like movement

### Question 2: Motion artifact separation

Can we distinguish clean physiological signal from motion-corrupted signal using:

- Accelerometer
- Gyroscope
- Heart-rate/PPG signal quality

- Personalized baselines

**Question 3: Stress detection feasibility**

Can we detect stress-related autonomic changes using personalized HRV/HR features compared against a resting baseline?

**Question 4: Tremor/movement detection**

Can we detect wrist movement patterns that resemble tremor or voluntary movement using the watch IMU?

**Question 5: Haptic cueing feasibility**

Can we deliver low-latency, synchronized haptic pulses on the watch for:

- Walking cadence cueing
- Breathing/stress pacing
- Tremor grounding experiments

## 3. Success Criteria for the Algorithmic Proof

You need objective pass/fail gates.

**A. Data collection success**

A session is usable if:

| Requirement | Target |
|---|---|
| Watch IMU data completeness | > 95% |
| Heart rate data completeness | > 90% |
| Reference ECG/chest strap data completeness | > 95% |
| Time synchronization error | < 100 ms after correction |
| Battery drain for a 60-minute protocol | Acceptable, ideally < 20% |
| Event labeling coverage | 100% of protocol blocks |

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**B. Heart rate / HRV validation**

Use an external ECG or chest strap as ground truth.

#### Resting state

| Metric | Target |
|---|---|
| Heart rate MAE vs ECG | < 3 bpm |
| HRV RMSSD correlation | > 0.85 |
| HRV RMSSD relative error | < 15% |
| Bad-segment detection sensitivity | > 85% |

#### Walking / movement state

| Metric | Target |
|---|---|
| Heart rate MAE vs ECG | < 6 bpm |
| HRV usable-window detection precision | > 85% |
| HRV usable-window detection recall | > 80% |

You may not get accurate full HRV during heavy movement immediately. That is fine. The first goal is to know **when the data is trustworthy**.

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### C. Stress detection feasibility

| Metric | Target |
|---|---|
| Stress task vs rest separation | Statistically significant within-subject change |
| Personalized stress classifier F1 | > 0.70 in initial healthy cohort |
| False trigger rate during quiet rest | < 2 events/hour |

Do not chase a universal stress model yet. Your first moat is **personalization**.

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### D. Tremor / abnormal movement detection

| Metric | Target |
|---|---|

| | |
|---|---|
| Tremor-like vs non-tremor movement classification F1 | > 0.80 |
| Tremor dominant frequency estimation error | < 0.5 Hz |
| False-positive rate during quiet rest | < 5% of epochs |

For Parkinson's tremor, the typical tremor band is often around **3–8 Hz**, commonly near **4–6 Hz**, but you should treat this as a parameter to learn from data.

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**E. Gait / walking cueing feasibility**

| Metric | Target |
|---|---|
| Step/cadence estimation MAE | < 5 steps/min |
| Gait vs non-gait detection F1 | > 0.90 |
| Haptic onset latency after trigger | < 150 ms |
| Haptic timing jitter | < 30 ms |
| Walking cadence cue usability | Subjective rating > 7/10 |

Freezing of gait detection is too advanced for the first proof. Start with:

- Gait detection
- Cadence estimation
- Gait irregularity
- Haptic cue synchronization

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## 4. Hardware Setup

**Primary device**

Samsung Galaxy Watch5 Pro R920 GPS

Expected relevant sensors:

- Optical heart rate / PPG-based heart rate
- Accelerometer

- Gyroscope
- Barometer
- GPS
- Haptic motor
- Possibly ECG/BIA features, but raw access may be restricted

Important assumption:

> Do not assume you will get raw PPG or raw ECG access from the watch as a third-party developer.

**Reference devices**

You need ground truth.

**Must-have**

1. **ECG chest strap**
   - Examples:
     - Polar H10
     - Polar Move ECG
     - Wellue ECG patch
     - Any BLE ECG device with exportable RR intervals or raw ECG
2. **Smartphone**
   - Android phone paired with the watch - Polar beat app
   - Companion app for logging, labeling, and storage
3. **Video recording**
   - For labeling movement, gait, tremor, and freezing-like events
   - Use a phone camera or webcam

**Nice-to-have**

1. Foot pod or shoe IMU for gait ground truth
2. GAITRite walkway or pressure mat if you have lab access
3. Respiratory belt for breathing ground truth
4. Metronome app for paced walking/breathing
5. Blood pressure cuff for baseline vitals
6. Standardized questionnaires for stress and symptom state

## 5. Software Setup

## Watch side

Build a Wear OS data-collection app.

#### Minimum watch app features

- Start/stop recording
- Sensor status check
- Battery logging
- Local storage if phone disconnects
- Periodic sync to phone
- Event markers
- Haptic cue engine
- Debug screen showing:
  - IMU sample rate
  - Heart rate availability
  - Battery
  - Storage
  - Connection state

#### Watch data sources

Try to collect:

| Data source | Priority |
|---|---|
| Accelerometer | High |
| Gyroscope | High |
| Heart rate | High |
| HRV if exposed | High |
| Raw PPG if accessible | Very high but uncertain |
| GPS | Medium |
| Battery level | Medium |
| Haptic event timestamps | High |

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## Phone side

Build a companion Android app.

#### Minimum phone app features

- Pair with watch
- Receive watch sensor streams
- Record reference ECG/chest strap data
- Record event labels
- Record session metadata
- Store files locally
- Export session package
- Show live signal quality for debugging

#### Phone data sources

| Data source | Priority |
|---|---|
| Chest strap ECG/RR intervals | High |
| Event annotations | High |
| Audio notes, optional | Low |
| GPS backup | Low |
| Session state machine | High |

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### Developer access to verify immediately

You need to check what Samsung allows.

#### Verify these items in week 1

1. Can your Wear OS app read continuous accelerometer and gyroscope data on the Watch5 Pro?
2. At what sample rate?
3. Can your app read heart rate continuously?
4. Can your app access HRV or RR intervals?
5. Can your app access raw PPG through Samsung Health SDK or Samsung Privileged Health SDK?
6. Can your app access ECG?

7. Can your app run reliably in the foreground for 60 minutes?
8. Can your app trigger custom haptic patterns with low latency?
9. Does Samsung Health sync data to Health Connect?
10. Can Health Connect provide retrospective HR/HRV data?

If raw PPG is unavailable, your algorithmic proof becomes:

> Can we build robust HRV quality scoring and stress inference from watch-derived HR/HRV plus IMU context?

That is still valuable.

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## 6. Test Cohorts

Do not start with Parkinson's patients. Start with a controlled pipeline.

### Phase 0: Founder pilot

**N = 1**

Purpose:

- Validate data pipeline
- Validate time sync
- Validate file formats
- Validate battery behavior
- Validate haptic latency

This is not for scientific proof.

### Phase 1: Healthy control cohort

**N = 10 to 20**

Purpose:

- Build motion-artifact detection
- Validate HR/HRV against ECG
- Train initial tremor-like movement detection
- Test stress task response
- Test walking cadence estimation

Include diversity where possible:

- Age range

- Skin tone
- Wrist size
- Arm hair
- Fitness level
- Baseline HRV variability

Why diversity matters: PPG quality can vary strongly with skin tone, perfusion, wrist size, and motion.

**Phase 2: Parkinson's pilot cohort**

Only after Phase 1 is stable.

**N = 5 to 12**

Purpose:

- Validate real PD tremor patterns
- Validate real gait abnormalities
- Validate stress response in PD population
- Test usability of haptic cues

Required:

- Ethical approval / IRB if collecting research data
- Clinical advisor
- Informed consent
- Fall risk protocol
- Safe testing environment
- Medical oversight

Do not run this phase alone in your apartment with no clinical supervision.

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## 7. Data Collection Protocol

Use the same protocol for each participant where possible.

Total time: **60–90 minutes**

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**Block 0: Consent and setup**

Duration: 10–15 minutes

Collect:

- Participant ID
- Age
- Sex
- Height
- Weight
- Hand dominance
- Watch wrist side
- Skin tone category, optional but useful for PPG bias analysis
- Medication state, if PD cohort
- PD severity, if PD cohort:
  - Hoehn & Yahr stage
  - MDS-UPDRS Part III subset if clinician available
- Current stress level, 0–10
- Current fatigue level, 0–10
- Caffeine intake
- Time since last meal
- Time since exercise

Do not collect identifying medical data unless you have proper consent and data governance.

---

### Block 1: Device placement and time synchronization

Duration: 5 minutes

#### Devices

Place:

- Watch on non-dominant wrist unless PD symptoms require otherwise
- Chest strap ECG
- Phone in pocket or armband
- Camera capturing full body and hands if possible

#### Time sync procedure

You need to synchronize watch, phone, and ECG clocks.

Do this at the beginning and end of every session.

#### Sync event 1: Triple tap

Tap the watch against the chest strap or phone three times.

#### Sync event 2: Five claps

Clap five times while all devices record.

#### Sync event 3: Sit-to-stand cycle

Perform 3 sit-to-stand movements.

These events will create spikes across accelerometer, ECG motion artifacts, video, and possibly heart-rate response.

Store:

- Phone timestamp
- Watch timestamp
- ECG timestamp
- Video timestamp if available

You will use these to estimate clock offset and drift.

### Block 2: Resting baseline

Duration: 5 minutes

Participant:

- Seated
- Arms supported
- Quiet
- No talking
- No phone use
- Normal breathing

Data goal:

- Clean baseline HRV
- Clean resting tremor detection baseline
- Low-motion signal quality reference

Event labels:

- `REST_START`
- `REST_END`

---

### Block 3: Paced breathing

Duration: 3 minutes

Use a metronome or visual breathing guide.

Suggested pattern:

- Inhale 4 seconds
- Exhale 6 seconds

This gives 6 breaths per minute, which is useful for HRV resonance and stress-pacing experiments.

Event labels:

- `PACED_BREATHING_START`
- `PACED_BREATHING_END`

Optional:

- Record respiratory belt if available.

---

### Block 4: Cognitive stress task

Duration: 5–7 minutes

Purpose:

Create measurable sympathetic activation without requiring medical stressors.

#### Recommended task

Mental arithmetic with social pressure.

Examples:

- Count backwards from 2026 by 7
- If wrong, restart
- Researcher gives neutral but firm feedback
- Or use Stroop test app

Do not use painful or unsafe stressors.

Collect:

- Self-reported stress before and after, 0–10
- Task start/end labels

Event labels:

- `STRESS_PRE_SCORE`
- `STRESS_TASK_START`
- `STRESS_TASK_END`
- `STRESS_POST_SCORE`

---

**Block 5: Movement artifact matrix**

Duration: 10 minutes

This block is critical. You need to teach your algorithm what bad signal looks like.

Have the participant perform each movement for 30–60 seconds with event labels.

| Task | Label |
|---|---|
| Still hand on lap | `MOTION_STILL` |
| Wrist flexion/extension | `MOTION_WRIST_FLEX` |
| Hand shaking 3–6 Hz | `MOTION_TREMOR_LIKE` |
| Finger tapping | `MOTION_TAPPING` |
| Writing gesture | `MOTION_WRITING` |
| Phone scrolling | `MOTION_SCROLLING` |
| Arm swing while standing | `MOTION_ARM_SWING` |
| Walking in place | `MOTION_WALK_IN_PLACE` |
| Hand rubbing | `MOTION_RUBBING` |
| Coughing/speaking movement | `MOTION_SPEECH_GESTURE` |

For PD participants, do not ask them to fake tremor if it is uncomfortable. For healthy participants, use instructed tremor-like shaking.

---

**Block 6: Tremor / hand motor tasks**

Duration: 10 minutes

If PD participant, this should be guided by a clinician.

**Task 1: Rest tremor**

- Seated
- Hands resting on thighs
- 60 seconds

Label:

- `TREMOR_REST_START`
- `TREMOR_REST_END`

#### Task 2: Postural tremor

- Arms extended forward
- 30–60 seconds

Label:

- `TREMOR_POSTURAL_START`
- `TREMOR_POSTURAL_END`

#### Task 3: Finger-to-nose

- 30 seconds per side

Label:

- `TREMOR_FTN_START`
- `TREMOR_FTN_END`

#### Task 4: Finger tapping

- Rapid finger tapping for 30 seconds

Label:

- `TREMOR_TAP_START`
- `TREMOR_TAP_END`

#### Task 5: Spiral drawing or handwriting, optional

- Draw a spiral or write a sentence
- Useful for upper-limb motor analysis

Label:

- `TREMOR_DRAW_START`
- `TREMOR_DRAW_END`

Video is very important here.

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### Block 7: Gait protocol

Duration: 15–20 minutes

Safety first. For PD participants, use a safe corridor, caregiver/clinician present, and stop if fall risk is high.

#### Task 1: Normal walking

- Walk 10 meters back and forth for 2–3 minutes
- Or treadmill at comfortable speed if available

Label:

- `GAIT_NORMAL_START`
- `GAIT_NORMAL_END`

#### Task 2: Slow walking

- 1 minute

Label:

- `GAIT_SLOW_START`
- `GAIT_SLOW_END`

#### Task 3: Fast walking

- 1 minute if safe

Label:

- `GAIT_FAST_START`
- `GAIT_FAST_END`

#### Task 4: Turning

- 180-degree turns
- 10 turns if safe

Label:

- `GAIT_TURN_START`
- `GAIT_TURN_END`

#### Task 5: Dual-task walking

- Walk while counting backwards
- Or walk while naming animals
- Only if safe

Label:

- `GAIT_DUAL_TASK_START`
- `GAIT_DUAL_TASK_END`

#### Task 6: Doorway/narrow space, optional

- Walk through doorway or marked narrow lane
- Useful for freezing-like phenomena in PD

Label:

- `GAIT_DOORWAY_START`
- `GAIT_DOORWAY_END`

#### Task 7: Freezing episode, if naturally occurring

Do not intentionally provoke unsafe freezing.

If freezing occurs naturally:

- Mark immediately
- Record video timestamp
- Have clinician annotate later

Label:

- `FOG_EVENT`

---

### Block 8: Haptic cueing test

Duration: 10 minutes

This tests whether the watch can deliver useful synchronized cues.

#### Test 1: Haptic latency

Create a test where the watch detects an event and triggers vibration.

Example:

- Participant taps watch face
- Watch logs timestamp
- Watch vibrates
- Phone/video records the event
- Measure delay

Label:

- `HAPTIC_LATENCY_TEST`

Target:

- Event to vibration < 150 ms

---

#### Test 2: Walking metronome

While walking:

- Estimate participant cadence for 30 seconds
- Deliver haptic pulses at:
  - 100% of cadence
  - 105% of cadence, optional
  - 90% of cadence, optional

Label:

- `HAPTIC_GAIT_CUE_START`
- `HAPTIC_GAIT_CUE_END`

Ask participant:

- Was the cue helpful?
- Was it annoying?
- Did it feel synchronized?
- Rating 0–10

---

#### Test 3: Breathing pacing

While seated:

- Deliver haptic pattern for 6 breaths/min
- Compare with guided breathing without haptics

Label:

- `HAPTIC_BREATHING_START`
- `HAPTIC_BREATHING_END`

Ask participant:

- Was the haptic rhythm clear?
- Did it help slow breathing?
- Rating 0–10

#### Test 4: Tremor grounding cue, exploratory

Do not claim this improves tremor.

While participant has tremor or tremor-like movement:

- Deliver low-frequency grounding pulses
- Example:
  - 10 ms pulse
  - 2–3 Hz
  - Gentle intensity

Label:

- `HAPTIC_TREMOR_CUE_START`
- `HAPTIC_TREMOR_CUE_END`

Collect subjective feedback only:

- Did it feel grounding?
- Did it distract?
- Did it increase awareness?
- Was it comfortable?

This is not therapeutic validation. It is usability and signal-trigger testing.

### Block 9: Recovery and questionnaires

Duration: 5 minutes

Collect:

- Post-session stress score
- Fatigue score
- Comfort score
- Watch comfort
- Haptic comfort
- Perceived usefulness
- Adverse symptoms, if any

Event labels:

- `SESSION_END`

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## 8. Event Label Dictionary

Use a consistent event taxonomy.

### Session events

| Event | Meaning |
|---|---|
| `SESSION_START` | Session begins |
| `DEVICE_SETUP_DONE` | Devices placed |
| `SYNC_TAPS` | Sync taps performed |
| `SYNC_CLAPS` | Sync claps performed |
| `SYNC_SIT_STAND` | Sit-to-stand sync performed |
| `SESSION_END` | Session ends |

### State events

| Event | Meaning |
|---|---|
| `REST_START` | Rest begins |
| `REST_END` | Rest ends |
| `STRESS_TASK_START` | Stress task begins |
| `STRESS_TASK_END` | Stress task ends |
| `PACED_BREATHING_START` | Paced breathing begins |
| `PACED_BREATHING_END` | Paced breathing ends |

### Motion events

| Event | Meaning |
|---|---|
| `MOTION_STILL` | No motion |
| `MOTION_WRIST_FLEX` | Wrist flexion/extension |
| `MOTION_TREMOR_LIKE` | Tremor-like shaking |
| `MOTION_TAPPING` | Tapping |

| | |
|---|---|
| `MOTION_SCROLLING` | Phone scrolling |
| `MOTION_ARM_SWING` | Arm swing |
| `MOTION_WALK_IN_PLACE` | Walking in place |

### Tremor events

| Event | Meaning |
|---|---|
| `TREMOR_REST_START` | Rest tremor task |
| `TREMOR_POSTURAL_START` | Postural tremor task |
| `TREMOR_FTN_START` | Finger-to-nose task |
| `TREMOR_TAP_START` | Finger tapping task |

### Gait events

| Event | Meaning |
|---|---|
| `GAIT_NORMAL_START` | Normal walking |
| `GAIT_SLOW_START` | Slow walking |
| `GAIT_FAST_START` | Fast walking |
| `GAIT_TURN_START` | Turning |
| `GAIT_DUAL_TASK_START` | Dual-task walking |
| `GAIT_DOORWAY_START` | Doorway walking |
| `FOG_EVENT` | Freezing-like event |

### Haptic events

| Event | Meaning |
|---|---|
| `HAPTIC_TRIGGER` | Trigger generated |
| `HAPTIC_ONSET` | Vibration started |
| `HAPTIC_OFFSET` | Vibration ended |

| | |
|---|---|
| `HAPTIC_GAIT_CUE_START` | Gait cueing starts |
| `HAPTIC_BREATHING_START` | Breathing cueing starts |
| `HAPTIC_TREMOR_CUE_START` | Tremor cueing starts |

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## 9. File Structure and Data Schema

Use a clean, immutable data layout.

### Directory structure

```
/data
  /raw
    /subject_001
      /session_001
        watch_imu.parquet
        watch_hr.parquet
        watch_haptics.parquet
        watch_battery.parquet
        phone_events.csv
        ecg_reference.parquet
        video_metadata.json
        session_metadata.json
        checksums.json

  /processed
    /subject_001
      /session_001
        synced_timeseries.parquet
        windows.parquet
        features.parquet
        quality_report.html

  /labels
    /subject_001
      /session_001
        manual_annotations.csv
        clinician_annotations.csv
```

---

### session_metadata.jso

```
{
  "subject_id": "sub-001",
  "session_id": "sess-001",
  "device_model": "Samsung Galaxy Watch5 Pro R920",
  "firmware_version": "unknown",
  "watch_side": "left",
  "phone_model": "Pixel 8",
  "ecg_device": "Polar H10",
  "session_start_epoch_ms": 1760000000000,
  "session_end_epoch_ms": 1760003600000,
  "protocol_version": "v0.1",
  "cohort": "healthy_control",
  "clinician_present": false,
  "notes": ""
}
```

**watch_imu.parquet**

| Column | Type | Description |
|---|---|---|
| `epoch_ms` | int64 | Converted timestamp |
| `elapsed_ms` | int64 | Device elapsed time |
| `ax` | float | Acceleration x |
| `ay` | float | Acceleration y |
| `az` | float | Acceleration z |
| `gx` | float | Gyro x |
| `gy` | float | Gyro y |
| `gz` | float | Gyro z |
| `motion_intensity` | float | Optional online feature |

**watch_hr.parquet**

| Column | Type | Description |
|---|---|---|
| `epoch_ms` | int64 | Timestamp |
| `hr_bpm` | float | Heart rate |
| `hrv_rmssd_ms` | float | If available |
| `hr_quality` | float | If available |
| `source` | string | Samsung Health / sensor |

**ecg_reference.parquet**

| Column | Type | Description |
|---|---|---|
| `epoch_ms` | int64 | Timestamp |

| | | |
|---|---|---|
| `ecg_raw` | float | If raw ECG available |
| `rr_ms` | float | RR interval |
| `hr_bpm` | float | ECG-derived HR |
| `beat_quality` | float | Quality flag |

---

**phone_events.csv**

| Column | Type | Description |
|---|---|---|
| `epoch_ms` | int64 | Timestamp |
| `event_name` | string | Event label |
| `source` | string | manual/auto |
| `notes` | string | Optional |

---

**manual_annotations.csv**

| Column | Type | Description |
|---|---|---|
| `start_epoch_ms` | int64 | Segment start |
| `end_epoch_ms` | int64 | Segment end |
| `activity` | string | rest/walk/tremor |
| `tremor_present` | boolean | Tremor label |
| `fog_present` | boolean | Freezing label |
| `stress_level` | float | 0–10 |
| `signal_quality` | float | good/bad |
| `annotator_id` | string | Annotator |

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## 10. Time Synchronization Plan

This is one of the most important parts.

If your timestamps are bad, your algorithmic proof is worthless.

### Required timestamp fields

For every sensor sample, store:

- `epoch_ms`
- `elapsed_ms`
- `source_device`
- `sample_index`

### Sync method

#### Step 1: NTP sync phone

Before session:

- Ensure phone time is network-synced.

#### Step 2: Watch-phone sync event

At session start:

- Press a button on phone app: `SYNC_START`
- Simultaneously tap watch three times.
- Log:
  - phone time
  - watch time

#### Step 3: ECG sync event

- Clap five times.
- Chest strap motion/artifact and video will show event.
- If ECG raw is available, use motion artifact or manual annotation.

#### Step 4: Cross-correlation

Use synchronized movement events to estimate offset:

```
offset = argmax cross_correlation(watch_motion, reference_motion)
```

#### Step 5: Drift correction

If session is longer than 10 minutes, use sync events at start and end to estimate linear clock drift.

```
corrected_time = raw_time + offset + drift * elapsed_time
```

### Acceptance criteria

After correction:

- Sync error < 100 ms for IMU events
- HR/HRV windows aligned to ECG windows within 250 ms
- Haptic event timestamps aligned to log within 50 ms locally

---

## 11. Data Processing Library Architecture

Build a Python library first. Do not overbuild the mobile UI.

Suggested name: daedalus_watch_kit

### Package structure

```
  tests/
    test_filters.py
    test_hrv.py
    test_sync.py
```

---

## 12. Core Processing Pipeline

### Stage 1: Ingestion

Load:

- Watch IMU
- Watch HR/HRV
- ECG/RR reference
- Event labels
- Manual annotations

Output:

- Standardized dataframes
- Session metadata
- Data completeness report

---

### Stage 2: Time synchronization

Tasks:

- Convert timestamps
- Estimate clock offsets

- Correct drift
- Align watch, phone, and ECG streams

Output:

- Unified timeline

---

### Stage 3: Windowing

Create fixed-length windows.

Recommended windows:

| Purpose | Window length | Overlap |
|---|---|---|
| Signal quality | 10 sec | 50% |
| HRV dynamic | 60 sec | 50% |
| HRV standard | 5 min | 0% |
| Tremor detection | 2 sec | 50% |
| Gait detection | 4 sec | 50% |
| Stress inference | 30–60 sec | 50% |

Each window should include:

- `subject_id`
- `session_id`
- `window_start_ms`
- `window_end_ms`
- `activity_label`
- `sensor_quality_flags`
- Features
- Target labels

---

### Stage 4: Signal quality scoring

You need a signal-quality index for every window.

#### For heart-rate/PPG quality

If raw PPG available:

Features:

- Pulse morphology quality
- Peak regularity
- Signal-to-noise ratio
- Entropy
- Motion intensity correlation
- Amplitude stability
- Missing sample ratio

If only HR available:

Features:

- HR variability plausibility
- Sudden HR jumps
- Missing HR samples
- IMU motion intensity
- Activity context

Output:

```
HR_SQI = 0 to 1
```

Where:

- `> 0.8` = good
- `0.5 to 0.8` = uncertain
- `< 0.5` = bad

#### For IMU quality

Features:

- Missing samples
- Clipping
- Sensor noise floor
- Unrealistic values
- Battery-saving mode indicators

Output:

```
IMU_SQI = 0 to 1
```

## 13. HRV Processing Plan

### If you have raw PPG

Pipeline:

1. Bandpass filter PPG
   - Example: 0.5–8 Hz
2. Remove baseline wander
3. Use accelerometer as reference for adaptive motion cancellation
4. Detect pulse peaks
5. Calculate inter-beat intervals
6. Remove ectopic/artifact beats
7. Interpolate IBI series
8. Compute HRV features
9. Compare against ECG-derived IBI

### If you only have HR or HRV estimates

Pipeline:

1. Use watch HR/HRV as source
2. Use IMU to identify motion-contaminated windows
3. Validate against ECG during clean windows
4. Build a quality classifier that says when watch HRV is usable
5. Use only high-quality windows for stress inference

This is still a strong algorithmic proof.

---

### HRV features to compute

#### Time domain

| Feature | Notes |
|---|---|
| `mean_hr` | Average heart rate |
| `sdnn` | Standard deviation of NN intervals |
| `rmssd` | Parasympathetic proxy |
| `pnn50` | Percentage of successive differences > 50 ms |

| | |
|---|---|
| `nn_outlier_rate` | Artifact/ectopy indicator |

#### Frequency domain

Use 5-minute windows where possible.

| Feature | Notes |
|---|---|
| `lf_power` | Low frequency |
| `hf_power` | High frequency |
| `lf_hf_ratio` | Use cautiously |
| `total_power` | Overall variability |

For irregular sampling, use:

- Lomb-Scargle periodogram
- Or resample IBI at 4 Hz then Welch

#### Nonlinear

| Feature | Notes |
|---|---|
| `sample_entropy` | Complexity |
| `dfa_alpha1` | Optional |
| `hrv_triangular_index` | Optional |

---

## 14. Stress Detection Plan

Do not build a universal stress model first.

Build a **personalized deviation model**.

### Step 1: Personal baseline

For each subject:

- Use 5-minute resting baseline
- Compute baseline:
  - Mean HR

- RMSSD
- SDNN
- LF/HF
- Sample entropy

**Step 2: Rolling features**

Compute features every 30–60 seconds.

**Step 3: Stress score**

Start with a simple rule-based score:

```
stress_score =
  w1 * z_score(HR)
  - w2 * z_score(RMSSD)
  + w3 * z_score(LF_HF)
  + w4 * motion_context_penalty
  - w5 * signal_quality_weight
```

Then normalize to 0–100.

**Step 4: Classifier**

Once enough data exists:

Features:

- HR z-score
- RMSSD z-score
- SDNN z-score
- LF/HF z-score
- Sample entropy
- Respiration proxy if available
- Motion intensity
- Time since rest
- Activity context

Labels:

- Stress task periods = stress
- Rest periods = non-stress
- Exclude motion-heavy periods initially

Models:

- Logistic regression first

- Random forest
- XGBoost
- Later: subject-adaptive models

### Evaluation

Use leave-one-session-out and leave-one-subject-out validation.

Targets:

- Within-subject F1 > 0.75
- Cross-subject F1 > 0.65 initially
- False alarm rate during rest < 2/hour

---

## 15. Tremor Detection Plan

### Goal

Detect wrist movement patterns consistent with tremor or rhythmic involuntary movement.

Do not claim clinical tremor severity yet.

### Input

- Accelerometer
- Gyroscope
- Optional heart-rate context
- Activity label

### Preprocessing

1. Remove gravity using low-pass filter
2. Compute acceleration magnitude
3. Compute angular velocity magnitude
4. Bandpass filter tremor band
   - Initial: 3–8 Hz
5. Compute short-time Fourier transform

Example:

```
window = 2 seconds
sampling_rate = 50 Hz
fft_size = 128 or 256
overlap = 50%
```

### Features

| Feature | Description |
|---|---|
| `band_power_3_8hz` | Power in tremor band |
| `dominant_freq` | Peak frequency |
| `spectral_entropy` | Regularity |
| `peak_prominence` | Strength of peak |
| `rms_accel` | Overall movement intensity |
| `rms_gyro` | Rotational movement intensity |
| `jerk` | Change in acceleration |
| `posture_context` | Rest/postural from orientation |
| `activity_context` | walking, typing, etc. |

### Classification

Start with rules:

```
tremor_candidate =
  band_power_3_8hz > threshold
  AND dominant_freq between 3 and 8 Hz
  AND spectral_entropy < threshold
  AND duration > 2 seconds
  AND not explained by walking/typing label
```

Then move to ML:

- Random forest
- XGBoost
- Small temporal model if enough data

### Output

For each 2-second window:

```
tremor_probability = 0 to 1
tremor_frequency_hz = float
tremor_amplitude_score = float
```

---

## 16. Gait Detection and Cadence Plan

### Goal

Detect walking and estimate cadence from wrist motion.

### Input

- Accelerometer
- Gyroscope
- GPS optional
- Barometer optional

### Preprocessing

1. Remove gravity
2. Compute acceleration magnitude
3. Bandpass for walking frequency
    - Initial: 0.5–3 Hz for general gait
    - Cadence may appear around 1–3 Hz at wrist depending on arm swing

### Step detection

Options:

1. Peak detection on filtered acceleration magnitude
2. Autocorrelation for cadence
3. Zero-crossing rate
4. Machine-learning step detector

### Cadence estimation

For each 4-second window:

- Compute autocorrelation
- Identify dominant period
- Convert to steps/min

```
cadence_steps_per_min = 60 / dominant_period_seconds
```

### Gait regularity features

| Feature | Description |
|---|---|
| `cadence` | Steps/min |
| `cadence_variability` | Variability across windows |
| `step_regularity` | Autocorrelation strength |
| `harmonic_ratio` | Smoothness |

| | |
|---|---|
| `arm_swing_amplitude` | Acceleration amplitude |
| `gait_bout_length` | Duration of continuous walking |

### Freezing proxy, early stage

Do not attempt clinical FoG detection immediately.

Instead detect:

- Sudden reduction in arm swing
- Increased cadence variability
- High-frequency freezing-like components if visible
- Gait hesitation patterns
- Contextual events: turning, doorway, dual-task

Label these as: gait_instability_candidate

Not definitive freezing.

---

## 17. Haptic Engine Test Plan

This is important because your product concept depends on synchronized vibrations.

### Haptic patterns to test

#### 1. Gait metronome

```text
```

```
```

```

1
2
3
```

```
```

```
Pulse interval = 60 / target_cadence
Pulse duration = 20 ms
Intensity = medium
```

Example:

- Cadence 100 steps/min
- Interval = 600 ms

#### 2. Breathing pacing

```
text
```

```

```

```

1
2
3
```

```

```

```
Inhale cue: longer or stronger pulse
Exhale cue: shorter or softer pulse
Pattern: 4 sec inhale, 6 sec exhale
```

#### 3. Tremor grounding

Exploratory only.

```
text
```

```

```

```

1
2
3
```

```

```

```
Pulse frequency: 2–3 Hz
Pulse duration: 10–20 ms
Intensity: gentle
```

### Latency test

#### Method

1. App records trigger event timestamp
2. Watch receives trigger
3. Watch starts vibration
4. Watch records haptic onset timestamp
5. If possible, use video/audio to verify physical onset

Metrics:

- Trigger-to-onset latency
- Onset jitter
- Missed pulse rate
- Battery impact

Targets:

| Metric | Target |
| --- | --- |
| Trigger-to-onset | < 150 ms |
| Jitter | < 30 ms |
| Missed pulses | < 1% |

---

## 18. Validation and Reporting Plan

For each participant session, generate a validation report.

### Report contents

#### 1. Session summary

- Duration
- Data completeness
- Battery drain
- Sensor availability
- Sync error

#### 2. Heart rate validation

- Watch HR vs ECG HR scatter plot
- Bland-Altman plot
- MAE by activity state

#### 3. HRV validation

- Watch RMSSD vs ECG RMSSD
- Correlation
- Bland-Altman
- Usable window detection performance

#### 4. Signal quality model

- Precision/recall for bad-segment detection
- Confusion matrix
- Feature importance

#### 5. Stress response

- Rest vs stress task HR/HRV distributions

- Within-subject z-score plots
- Classifier performance

#### 6. Tremor detection

- Spectrogram examples
- Confusion matrix
- Dominant frequency estimates
- False-positive analysis

#### 7. Gait detection

- Cadence error
- Gait bout detection
- Turn detection, if available
- Haptic synchronization during walking

#### 8. Haptic performance

- Latency histogram
- Jitter
- Subjective ratings

---

## 19. Six-Week Execution Plan

### Week 1: Device bring-up and data access

Goals:

- Receive watch
- Set up Android Studio
- Set up Wear OS app
- Pair watch with phone
- Enable developer mode
- Test sensor access
- Test heart-rate access
- Test Samsung Health / Health Connect access
- Test haptic engine
- Order/reference ECG chest strap

Deliverables:

- Sensor access matrix
- Sample-rate report
- Raw data capture demo
- Haptic latency spike test

Go/no-go:

- If accelerometer/gyro continuous access works, proceed.
- If heart-rate access is limited, plan fallback using Health Connect and external ECG.

---

**Week 2: Data pipeline MVP**

Goals:

- Build ingestion pipeline
- Build file schema
- Build event logger
- Build time sync module
- Run one founder session
- Export data to Python

Deliverables:

- One complete session folder
- Time-sync report
- First plots of IMU, HR, ECG, events

Go/no-go:

- If time sync error > 250 ms, do not proceed to modeling. Fix sync first.

---

**Week 3: Healthy pilot data collection**

Goals:

- Collect 3–5 healthy sessions
- Validate HR/HRV during rest and stress task
- Validate motion artifact matrix
- Build first signal-quality features

Deliverables:

- HR validation report

- HRV usable-window report
- Motion artifact feature plots

Go/no-go:

- If HR MAE during rest > 5 bpm, fix ingestion/sync before expanding.

---

**Week 4: Artifact and stress models**

Goals:

- Build HRV quality classifier
- Build personalized stress score
- Build first tremor-like movement detector
- Expand to 8–10 healthy participants

Deliverables:

- Artifact classifier v0.1
- Stress score v0.1
- Tremor feature notebook
- Updated validation report

Go/no-go:

- If artifact detection cannot separate rest from motion-heavy windows, revise feature set.

---

**Week 5: Gait and haptic cueing**

Goals:

- Build gait detection
- Build cadence estimation
- Test haptic metronome
- Test breathing pacing
- Collect 5 more healthy sessions

Deliverables:

- Cadence validation report
- Haptic latency report
- Gait cueing usability report

Go/no-go:

- If haptic latency > 300 ms, optimize trigger path before PD pilot.

---

**Week 6: Algorithmic proof report**

Goals:

- Consolidate results
- Build VC-facing proof report
- Identify PD pilot protocol
- Identify regulatory/ethics requirements
- Decide whether to pursue clinical partner

Deliverables:

- Final algorithmic proof report
- Dataset summary
- Model performance tables
- Error analysis
- Next-step plan
- Clinical partnership target list

Go/no-go for next phase:

- Proceed to PD pilot only if:
  - HR/HRV quality scoring works
  - Haptic latency is acceptable
  - Tremor/gait detection shows feasibility
  - Clinical advisor agrees to supervised pilot

---

## 20. Parkinson's Pilot Protocol Additions

Only run this after healthy cohort proof.

**Required**

- Clinical collaborator
- IRB/ethics approval if applicable
- Informed consent
- Fall prevention plan
- Caregiver/clinician present

- Safe environment
- Insurance/risk assessment if commercial pilot

### Additional PD data

For each PD participant:

- Diagnosis duration
- Medication state:
  - ON medication
  - OFF medication if safe and clinically appropriate
- Time since last dose
- Hoehn & Yahr stage
- MDS-UPDRS Part III if clinician available
- History of freezing of gait
- Fall history
- Tremor-dominant vs non-tremor-dominant symptoms

### Additional tasks

- Rest tremor assessment
- Postural tremor assessment
- Finger tapping
- Hand movements
- Pronation/supination
- Gait with turns
- Dual-task gait
- Doorway walking
- Speech task, optional

### Speech task, optional

Since you mentioned talking:

#### Task

Participant reads a standard passage for 60 seconds.

Conditions:

1. No haptic cue
2. Haptic pacing cue

If audio is recorded:

- Measure speaking rate
- Pause duration
- Speech variability

If audio is not recorded:

- Collect subjective ratings only

Label:

- `SPEECH_BASELINE_START`
- `SPEECH_BASELINE_END`
- `SPEECH_HAPTIC_START`
- `SPEECH_HAPTIC_END`

Do not claim speech improvement yet.

---

## 21. Data Quality Rules

**Reject session if**

- Watch disconnected for > 10% of session
- ECG reference missing for > 10% of session
- Time sync cannot be corrected to < 250 ms
- Event labels missing for major protocol blocks
- Battery failure or device overheating
- Participant safety issue

**Flag window if**

- Motion intensity exceeds sensor range
- HR values physiologically implausible:
  - < 30 bpm
  - 220 bpm
- RR intervals implausible:
  - < 300 ms
  - 2000 ms
- PPG amplitude flat
- Missing samples > 20%

- Haptic command not acknowledged

---

## 22. Privacy and Safety Requirements

### Privacy

- Use participant IDs only
- Do not store names in raw data files
- Keep consent forms separately
- Encrypt drives
- Use access controls
- Do not upload raw data to public repositories
- Use DVC or similar for data versioning

### Safety

- Stop immediately if participant feels dizzy, anxious, or unsafe
- Do not provoke freezing intentionally
- Do not test walking tasks near stairs
- Use a clinician for PD cohort
- Document adverse events
- Do not provide medical advice through the app

### Regulatory language

Use language like:

- “research prototype”
- “wellness logging”
- “movement pattern analysis”
- “rhythmic cueing”
- “data visualization”

Avoid language like:

- “treats Parkinson’s”
- “improves tremor”
- “prevents falls”
- “cures freezing”
- “medical therapy”

---

## 23. Main Risks and Mitigations

### Risk 1: No raw PPG access

Mitigation:

- Use HR/HRV estimates
- Use external ECG as ground truth
- Focus on motion-aware HRV quality scoring
- Apply for Samsung privileged health access if relevant

---

### Risk 2: Watch HRV inaccurate during tremor

Mitigation:

- Do not assume HRV is always valid
- Build a confidence score
- Use only high-confidence windows
- Use personalized baselines
- Fuse IMU context with HRV

---

### Risk 3: Time synchronization failure

Mitigation:

- Use physical sync events
- Log multiple clocks
- Validate with cross-correlation
- Reject bad sessions

---

### Risk 4: Battery drain too high

Mitigation:

- Reduce IMU sample rate to 25–50 Hz
- Batch sensor delivery
- Disable GPS when not needed
- Use foreground service efficiently
- Limit continuous recording to protocol blocks

---

### Risk 5: Haptic latency too high

Mitigation:

- Trigger haptics locally on watch when possible
- Avoid phone-to-watch round trip for time-critical cues
- Precompute haptic patterns
- Test different Android/Wear OS vibration APIs

---

**Risk 6: PD cohort recruitment difficulty**

Mitigation:

- Start with healthy cohort
- Partner with movement disorder clinic
- Partner with university motor control lab
- Use public PD datasets for pretraining
- Build clinician advisory relationship early

---

## 24. Minimum Viable Dataset Target

Before trying to raise money or recruit PD patients, aim for this dataset.

**Healthy cohort**

- 12 participants
- 1 session each
- 60 minutes each
- Total: ~12 hours

Each session must contain:

- 5 min rest
- 3 min paced breathing
- 5 min stress task
- 10 min movement artifact matrix
- 10 min tremor/hand tasks
- 15 min gait tasks
- 10 min haptic tests

**Derived dataset**

For each session:

- Clean synced time series
- Window-level feature table
- Quality labels
- ECG ground truth alignment
- Session validation report

That dataset becomes your algorithmic proof and early proprietary data moat.

---

## 25. Deliverable Package for Investors or Partners

When you finish the proof, package it like this:

**1. Technical proof deck**

- Problem
- Data collection method
- Signal pipeline
- Validation results
- Error analysis
- Roadmap

**2. Algorithmic proof report**

- HR/HRV accuracy
- Motion artifact detection
- Stress feasibility
- Tremor detection
- Gait/cadence detection
- Haptic latency

**3. Dataset summary**

- Number of participants
- Total hours
- Total windows
- Data completeness
- Cohort diversity
- Label coverage

**4. Demo**

- Live watch data collection
- Real-time signal quality display
- Haptic cueing demo
- Visualization dashboard

**5. Regulatory positioning**

- Wellness/research prototype
- No treatment claims
- Pathway to future clinical validation

**6. Next ask**

- Funding or partnership for PD pilot
- Clinical advisor recruitment
- IRB/ethics submission
- Letter of intent from clinic/CRO

## 26. Immediate Next 72 Hours

Do this first.

**Day 1**

- Set up Samsung developer account
- Install Android Studio
- Create Wear OS hello-world app
- Pair Watch5 Pro with phone
- Enable developer mode on watch
- Test accelerometer and gyroscope access

**Day 2**

- Test heart-rate sensor access
- Check Samsung Health SDK access
- Check Health Connect availability
- Order ECG chest strap if not already ordered
- Create session file structure

**Day 3**

- Build first data logger:
  - Accelerometer

    - Gyroscope
    - Heart rate
    - Event buttons
- Run a 10-minute founder test
- Plot the data
- Check timestamps and missing samples

---

## 27. The Single Most Important Metric to Prove First

Before anything else, prove this:

> You can generate a window-level signal-quality score that tells you when the watch-derived HR/HRV data is trustworthy during rest, stress, movement, and tremor-like motion.

If you can do that, you can build:

- Personalized stress detection
- Context-aware tremor detection
- Gait cueing triggers
- Haptic personalization
- Clinical data product

If you cannot do that, everything else is built on noise.

---

## 28. Final Test Plan Summary

**Build**

- Wear OS sensor logger
- Phone companion logger
- ECG reference pipeline
- Event annotation system
- Python processing library

**Collect**

- 1 founder pilot
- 12–20 healthy participants

- Later, 5–12 PD participants under supervision

**Process**

- Time sync
- Windowing
- Signal quality scoring
- HRV extraction
- Motion/tremor features
- Gait features
- Stress personalization
- Haptic latency analysis

**Validate**

- HR/HRV vs ECG
- Artifact detection
- Stress task separation
- Tremor-like movement detection
- Gait cadence estimation
- Haptic latency/jitter

**Deliver**

- Algorithmic proof report
- Proprietary dataset
- VC-ready technical moat
- Clinical pilot plan